**PUNJAB ENGINEERING COLLEGE (DEEMED TO BE UNIVERSITY)**
Sector 12, Chandigarh - 160012 | Office of Dean Academic Affairs (DAA)
**Capstone Coursework Deliverable • Academic Clearance Automation Engine**

# Data Flow Diagrams (DFD) & Data Dictionary

## Academic Grievance, No-Dues & Clearance Automation Engine

**Institution:** Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh - 160012
**Document Identifier:** PEC-SE-DFD-FROZEN
**Standard Compliance:** Structured Systems Analysis & Design (Yourdon & DeMarco / Gane & Sarson Standards)
**Baseline Version:** 3.0.0 (FROZEN ARCHITECTURAL BASELINE)
**Administrative Authority:** Office of the Dean Academic Affairs (DAA), PEC Chandigarh
**Status:** FROZEN & RATIFIED (Implementation-Agnostic Baseline)

### Baseline Freeze Declaration

> **NOTICE OF BASELINE FREEZE:**
> This document specifies the complete, immutable data transformations, data store structures, and information flow boundaries for the PEC Clearance & Grievance Automation Engine.
>
> The process hierarchies, data store schemas, input/output data flows, and data element dictionaries defined herein are **FROZEN**. Implementation changes (e.g., frontend state management, database engine replacement, or network transport layers) preserve these exact inputs, transformations, and outputs, ensuring zero modifications to this document.

## 1. Notational Conventions & Symbol Guide

| Symbol Element | Gane & Sarson / Yourdon Notation | Mermaid Representation | Semantic Meaning |
|---|---|---|---|
| **External Entity** | Double Square / Rectangle | `Entity["External Entity"]` | Source or destination of data outside system boundaries (e.g., Student, Staff). |
| **Process** | Rounded Rectangle / Circle | `Process(("ID: Process Name"))` | Transforms incoming data flows into outgoing data flows. |
| **Data Store** | Open Rectangle / Parallel Lines | `Store[("DS-ID: Data Store")]` | Repository of persistent data at rest (e.g., relational tables). |
| **Data Flow** | Directed Vector Arrow | `A -- "Data Flow Name" --> B` | Pipeline carrying packetized data elements between entities, processes, and stores. |

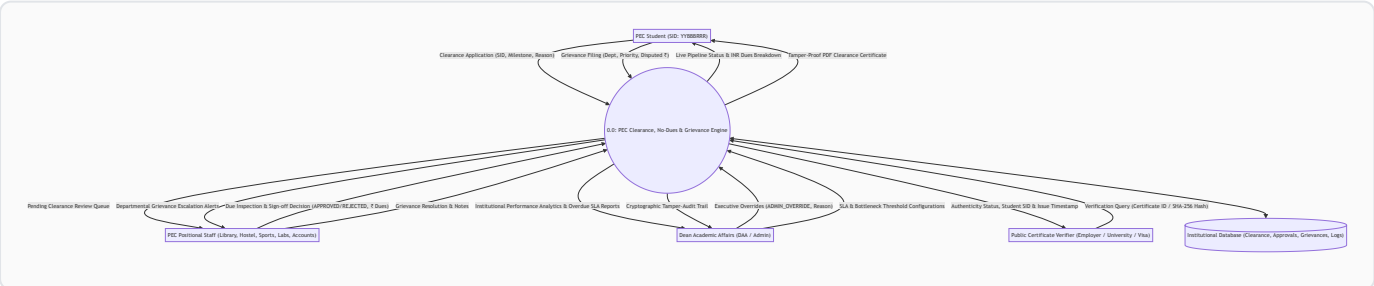
### 1.1 Formal DFD Construction Rules & Quality Compliance

This specification adheres strictly to the classic **Yourdon & DeMarco** and **Gane & Sarson** structured systems analysis rules:

| Structured Analysis Rule | Rule Description & Formal Constraint | Compliance Verification in this DFD |
|---|---|---|
| **Rule 1: Conservation of Data** | A process cannot create data out of nothing ("Miracle") nor consume data without producing output ("Black Hole"). | Every process ( `1.0` through `6.0` and `2.1` .. `5.4` ) transforms identifiable inputs into outputs. |
| **Rule 2: Entity-to-Entity Forbidden** | External entities cannot communicate directly within the DFD boundary without passing through a process. | Verified: All student, staff, and admin interactions pass through numbered processes. |
| **Rule 3: Store-to-Store Forbidden** | Data cannot move directly from one data store to another without process manipulation. | Verified: Table updates (e.g. `DS-4` to `DS-3` ) are mediated by processes `3.0` , `5.1` , and `5.2` . |
| **Rule 4: Entity-to-Store Forbidden** | External entities cannot directly read or write to data stores without an intervening process. | Verified: All persistence calls are strictly routed through processes `1.0` , `2.0` , `3.0` , `4.0` . |
| **Rule 5: Balancing Rule (Horizontal & Vertical)** | Inputs and outputs of a parent process at Level $N$ must exactly match the aggregate boundary flows at Level $N+1$. | Verified: Level 0 external flows balance with Level 1, and Level 1 processes balance with Level 2. |
| **Rule 6: Unique Hierarchical Numbering** | Every process is uniquely identified using hierarchical decimal notation ( `0.0` $\rightarrow$ `1.0` .. `6.0` $\rightarrow$ `2.1` .. `2.4` ). | Verified: Clear hierarchical numbering applied across all levels. |

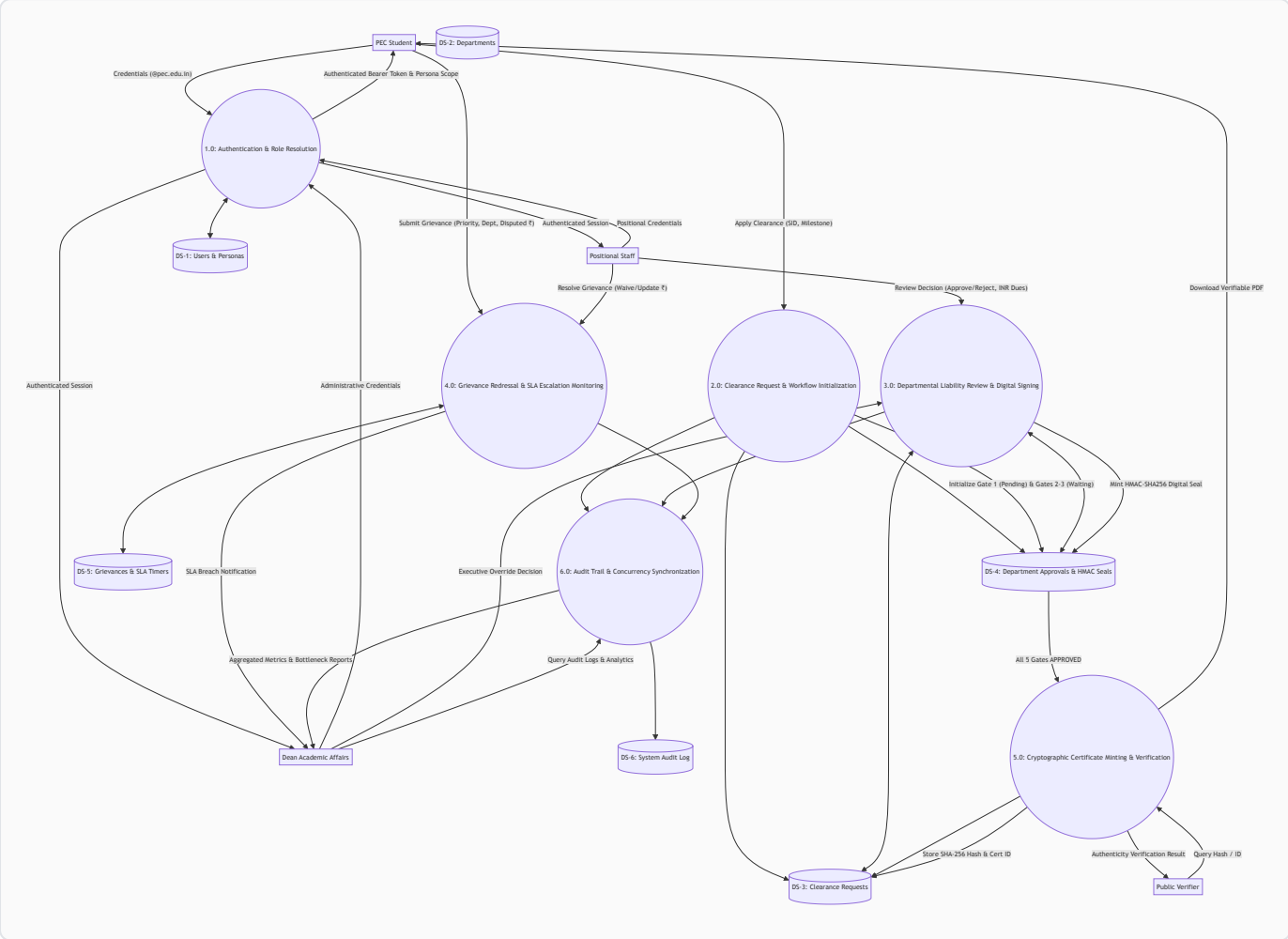
## 2. DFD Level 0: Context Analysis Diagram

The Level 0 Context Diagram establishes the highest-level operational boundary of the Clearance Engine within the Punjab Engineering College ecosystem. It identifies the four primary external entities interacting with the central clearance processing environment:



# 3. DFD Level 1: System Decomposition Diagram

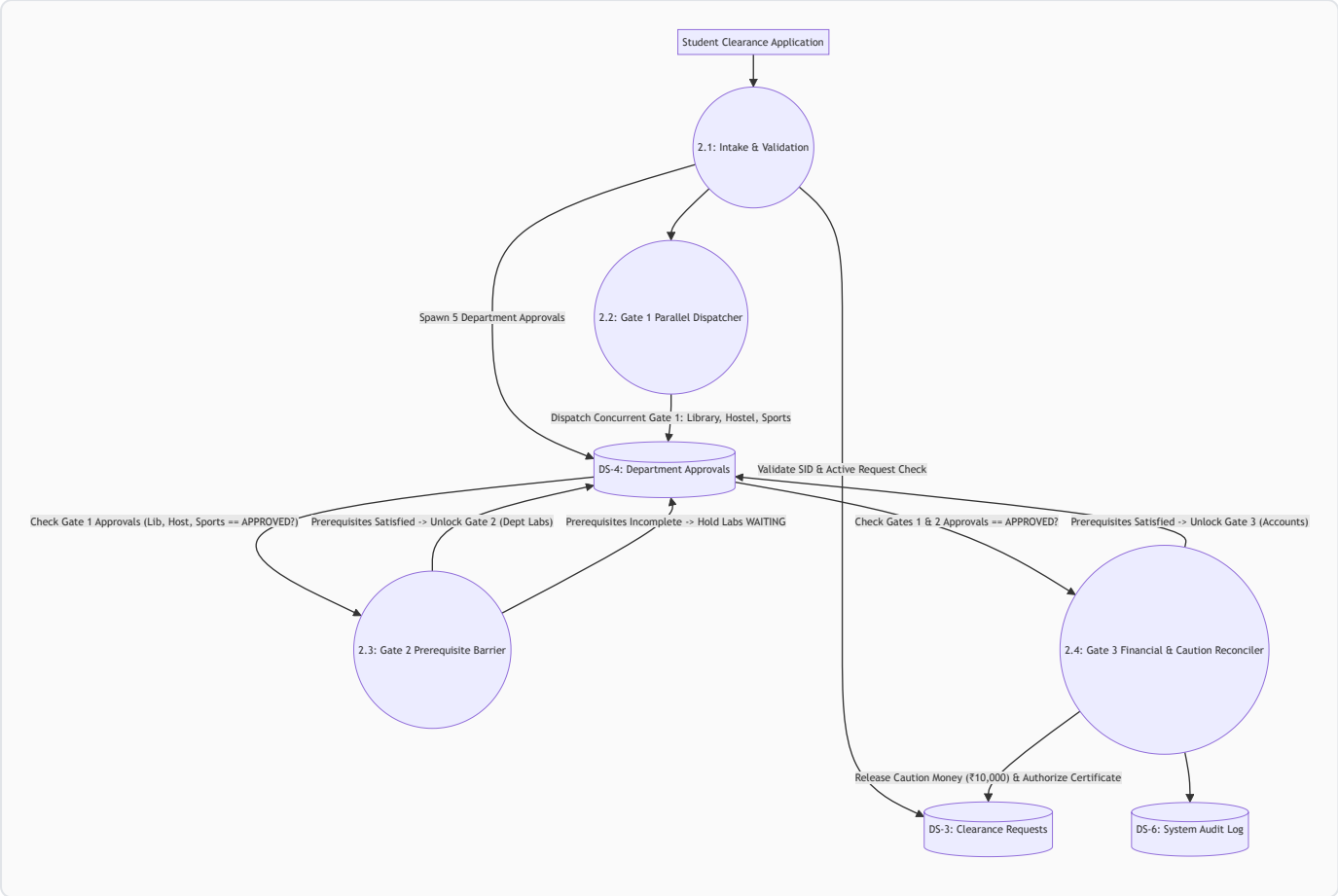
The Level 1 DFD decomposes the top-level process `0.0` into six modular, cohesive functional processes governing authentication, clearance workflow, liability assessment, grievance arbitration, cryptographic certification, and audit synchronization:




# 4. DFD Level 2: Sub-Process Decompositions

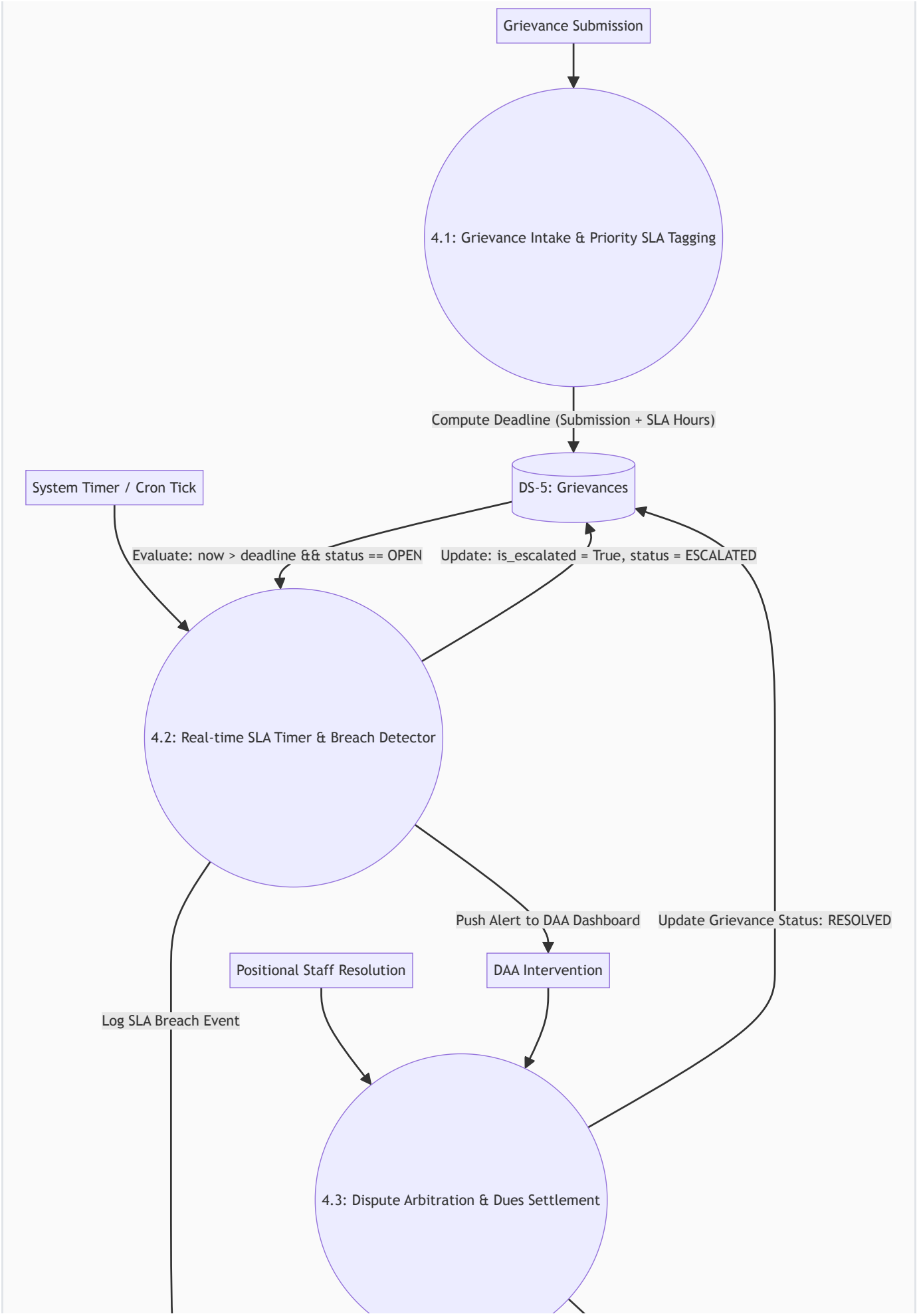
## 4.1 Process 2.0 Decomposition: Clearance Gating & Chain of Responsibility

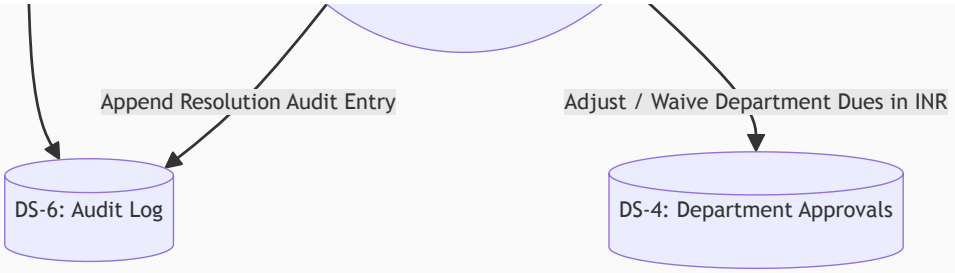
This diagram isolates how departmental clearance requests flow across the three sequential and parallel validation gates:




## 4.2 Process 4.0 Decomposition: Grievance Redressal & SLA Escalation Engine

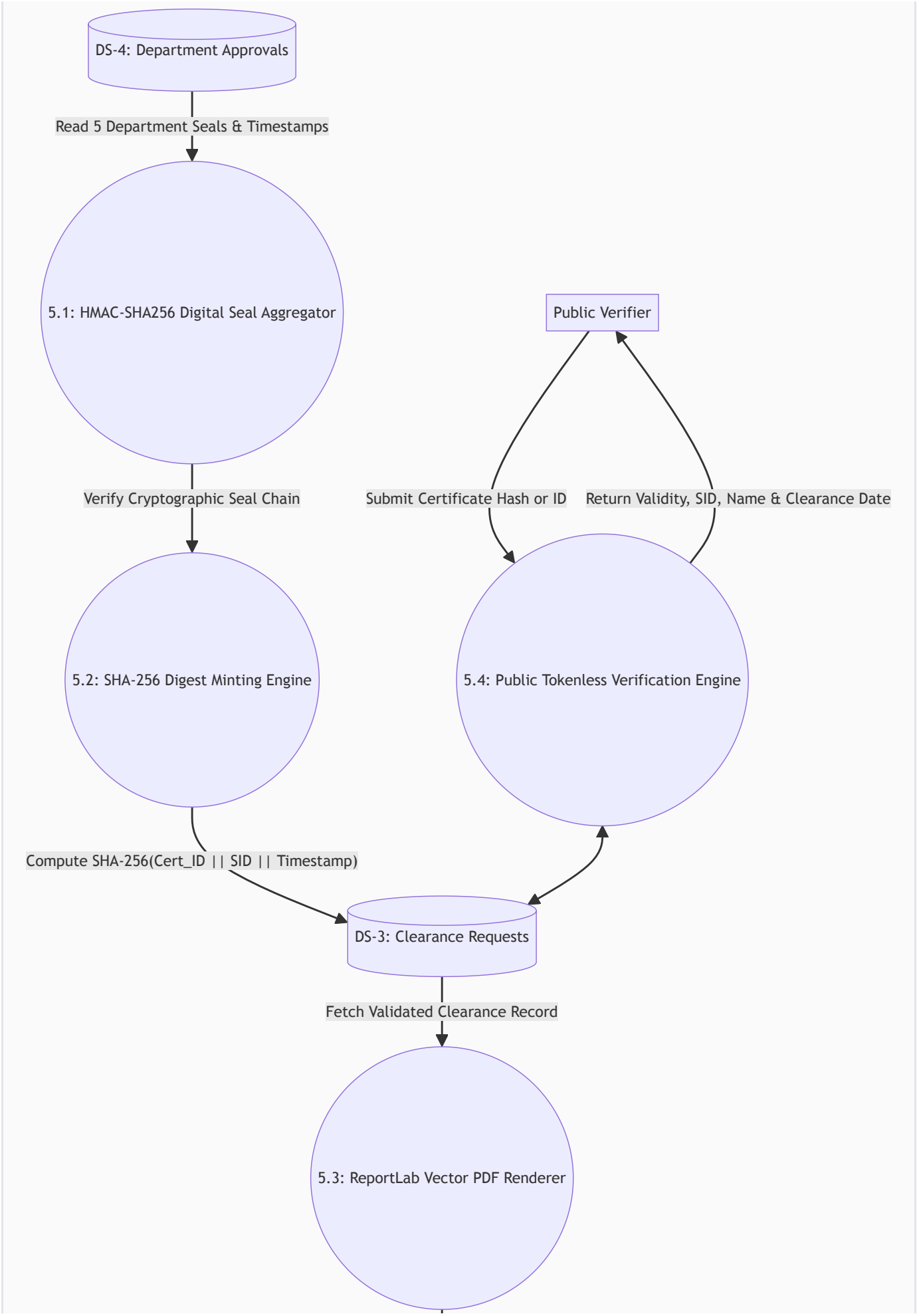
This diagram details the priority-based SLA calculation, automatic timer evaluation, and DAA escalation workflow:

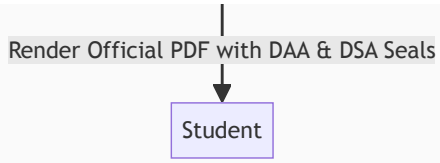

Grievance Submission
4.1: Grievance Intake & Priority SLA Tagging
Compute Deadline (Submission + SLA Hours)
DS-5: Grievances
System Timer / Cron Tick
Evaluate: now > deadline && status == OPEN
Update: is_escalated = True, status = ESCALATED
4.2: Real-time SLA Timer & Breach Detector
Push Alert to DAA Dashboard
Update Grievance Status: RESOLVED
Positional Staff Resolution
DAA Intervention
Log SLA Breach Event
4.3: Dispute Arbitration & Dues Settlement




## 4.3 Process 5.0 Decomposition: Cryptographic Certificate Minting & Public Verification

This diagram details the generation of HMAC digital signatures, SHA-256 certificate hashing, and unauthenticated public verification:


DS-4: Department Approvals
Read 5 Department Seals & Timestamps
5.1: HMAC-SHA256 Digital Seal Aggregator
Verify Cryptographic Seal Chain
5.2: SHA-256 Digest Minting Engine
Compute SHA-256(Cert_ID || SID || Timestamp)
DS-3: Clearance Requests
Fetch Validated Clearance Record
5.3: ReportLab Vector PDF Renderer
Public Verifier
Submit Certificate Hash or ID
Return Validity, SID, Name & Clearance Date
5.4: Public Tokenless Verification Engine




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# 5. Comprehensive Data Dictionary

## 5.1 Data Store Schemas (Data at Rest)

### `DS-1` : Users & Personas ( `users` table)

| Field Name | Type | Key | Description | Example |
|---|---|---|---|---|
| `id` | Integer | PK | Unique user record surrogate key | `1` |
| `email` | String(120) | Unique | Institutional PEC email address | `alice@pec.edu.in` |
| `full_name` | String(120) | - | Student or Positional Staff Name | `Alice Sharma` / `Chief Warden (Hostels)` |
| `role` | String(20) | - | Role scope ( `STUDENT` , `STAFF` , `ADMIN` ) | `STUDENT` |
| `sid` | String(20) | Nullable | 8-digit PEC Student Identification Number | `22103042` |
| `department_code` | String(20) | Nullable | Assigned department for staff | `HOSTEL` |
| `is_active` | Boolean | - | Account active status flag | `True` |

### `DS-2` : Departments ( `departments` table)

| Field Name | Type | Key | Description | Example |
|---|---|---|---|---|
| `id` | Integer | PK | Department record identifier | `2` |
| `code` | String(20) | Unique | Institutional department code | `HOSTEL` |
| `name` | String(120) | - | Official facility designation | `PEC Hostels (Shivalik/Kurukshetra/etc.)` |
| `head_title` | String(120) | - | Permanent administrative title | `Chief Warden (Hostels)` |
| `gate_number` | Integer | - | Pipeline gate assignment ( `1` , `2` , or `3` ) | `1` |

### `DS-3` : Clearance Requests (`clearance_requests` table)

| Field Name | Type | Key | Description | Example |
|---|---|---|---|---|
| `id` | Integer | PK | Clearance request identifier | `1` |
| `student_id` | Integer | FK | References `users.id` | `1` |
| `academic_year` | String(20) | - | Academic session | `Final Academic Session` |
| `semester` | Integer | - | Current enrolled semester | `8` |
| `reason` | String(120) | - | Clearance justification | `B.Tech Final Degree Graduation` |
| `status` | String(30) | - | Overall status ( `PENDING` , `APPROVED` , `REJECTED` ) | `APPROVED` |
| `certificate_id` | String(50) | Unique | Minted institutional certificate number | `PEC-CLR-YYYY-XXXXX` |
| `certificate_hash` | String(64) | Unique | Cryptographic SHA-256 certificate digest | `e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b` |

### `DS-4` : Department Approvals (`department_approvals` table)

| Field Name | Type | Key | Description | Example |
|---|---|---|---|---|
| `id` | Integer | PK | Approval gate record key | `1` |
| `request_id` | Integer | FK | References `clearance_requests.id` | `1` |
| `department_id` | Integer | FK | References `departments.id` | `2` |
| `status` | String(30) | - | `WAITING_PREREQUISITE`, `PENDING`, `APPROVED`, `REJECTED` | `APPROVED` |
| `due_amount` | Float | - | Outstanding department liability in INR (`₹`) | `3500.0` |
| `remarks` | Text | - | Positional reviewer notes | `Shivalik Mess Dues Reconciled` |
| `digital_signature` | String(128) | - | Cryptographic HMAC-SHA256 signature seal | `a8f9c1d2e3b4a5...` |
| `signed_at` | DateTime | - | Timestamp of cryptographic signing | `[Verification Timestamp]` |

### `DS-5`: Grievances (`grievances` table)

| Field Name | Type | Key | Description | Example |
|---|---|---|---|---|
| `id` | Integer | PK | Grievance ticket identifier | `1` |
| `student_id` | Integer | FK | References `users.id` | `1` |
| `department_id` | Integer | FK | References `departments.id` | `2` |
| `subject` | String(200) | - | Concise grievance description | `Incorrect Mess Dues for Internship Term` |
| `priority` | String(20) | - | SLA Priority (`CRITICAL`, `HIGH`, `MEDIUM`, `LOW`) | `HIGH` |
| `disputed_amount` | Float | - | Contested liability in INR (`₹`) | `3500.0` |
| `status` | String(20) | - | `OPEN`, `RESOLVED`, `REJECTED`, `ESCALATED` | `ESCALATED` |
| `sla_deadline` | DateTime | - | Dynamic computed resolution deadline | `[Calculated SLA Deadline]` |
| `is_escalated` | Boolean | - | Flag indicating DAA SLA breach escalation | `True` |

### `DS-6`: Audit Log (`audit_logs` table)

| Field Name | Type | Key | Description | Example |
|---|---|---|---|---|
| `id` | Integer | PK | Sequential audit event identifier | `101` |
| `user_id` | Integer | FK | Actor responsible for the state mutation | `4` |
| `action` | String(50) | - | Standard action verb | `APPROVAL_SIGNED`, `ADMIN_OVERRIDE` |
| `entity_type` | String(50) | - | Target domain entity | `ClearanceRequest`, `Grievance` |
| `details` | Text | - | JSON-encoded mutation metadata | `{"dept": "HOSTEL", "status": "APPROVED"}` |
| `timestamp` | DateTime | - | UTC/IST creation timestamp | `[Transaction Timestamp]` |

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## 5.2 Core Data Element Dictionary (Data in Motion)

1. **`SID` (Student Identification Number):**
   - **Type:** String (8 characters alphanumeric).
   - **Format:** `YYBBBRRR` where `YY` is enrollment year, `BBB` is academic branch code (`103` = CSE), and `RRR` is roll index.
   - **Scope:** Immutable institutional identifier.
2. **`due_amount_inr` (Financial Liability):**
   - **Type:** IEEE 754 Floating Point (formatted as `en-IN` decimal string in client).
   - **Currency:** Indian Rupee (`₹`).
   - **Allowable Range:** $[0.00, ; 500000.00]$.
3. **`digital_signature` (HMAC Seal):**
   - **Type:** Hexadecimal string (64 characters).
   - **Algorithm:** $\text{HMAC-SHA256}(K_{\text{institutional}}, \text{staff_id} \parallel \text{dept_code} \parallel \text{request_id} \parallel \text{timestamp})$.
4. **`certificate_hash` (Verification Digest):**
   - **Type:** Hexadecimal string (64 characters).
   - **Algorithm:** $\text{SHA-256}(\text{cert_id} \parallel \text{student_sid} \parallel \text{academic_year} \parallel \text{completion_time})$.
5. **`sla_priority`:**
   - **Type:** Enumeration (`CRITICAL`, `HIGH`, `MEDIUM`, `LOW`).
   - **Mapping:** `CRITICAL` $\rightarrow 4\text{h}$, `HIGH` $\rightarrow 12\text{h}$, `MEDIUM` $\rightarrow 24\text{h}$, `LOW` $\rightarrow 48\text{h}$.